

Potential to improve winter grazing pastures: Sieben Land and Livestock study

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On the Ground

- Maintaining economic sustainability requires reduced inputs such as mechanically harvested forage. It is estimated that grazing versus feeding cattle during the winter can save 42% to 70% of the yearly input costs in the western United States and Saskatchewan, Canada.
- Grass mixtures of intermediate wheatgrass and meadow brome grass produced 2 and 3 times the stockpiled forage than orchardgrass and native range, respectively.
- Economic pay-back period on the initial pasture establishment costs were < 1 year for intermediate wheatgrass, meadow brome grass/intermediate wheatgrass, and meadow brome grass/tall fescue mixtures.

Keywords: Economic sustainability, Winter grazing, Stockpiled forage, Quality, Intermediate wheatgrass, Meadow brome grass, Tall fescue, Orchardgrass

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Introduction

Many cow-calf operations rely heavily on grazed perennial forage (native range and improved pastures) throughout the growing season and have traditionally overwintered cows in confined feeding areas such as dry lots/pastures with hay or silage offered.¹ It is estimated that feeding over wintering cattle accounts for two-thirds of the costs associated with cow/calf production in western Canada.^{2,3} Saskatchewan Forage Council⁴ estimated a 42% savings for stockpiled grasses relative to a traditional confined winter-feeding strategy that relied heavily on mechanical harvesting/swathing.

Limiting factors associated with winter grazing include reduced forage availability, poor forage nutritional quality, absence of watering systems, extreme cold tolerance, potential wasted feed, animal health, reduced animal performance, winter calving, and depth and crusting of the snow.⁴ Consideration of these factors in addition to animal nutritional requirements and plant material options are essential to achieve success by extending the grazing system into the fall and winter.

Forage nutritional quality is perhaps one of the biggest constraints to grazing fall and winter stockpiled forage by livestock. As plants mature, there is an observed decrease in crude protein (CP) and fiber digestibility and increases in fiber concentrations.^{5,6} Jensen et al.⁷ reported CP values below 5% not meeting the recommended 7% to 9% for nonlactating cows in mid- to-late gestation from November to March on 14 range grass species.⁸

A study characterizing chemical composition of range plants, reported CP values declined in rough fescue (*Festuca campestris* Rydb.) from 6.6% to 4.2% from late summer to post winter, and in vitro dry-matter (DM) digestibility declined from 36.3% to 24.0% for the same time period.⁹ A similar decline in forage quality was observed by Willms¹⁰ for Altai wildrye (*Leymus angustus* [Trin.] Pilger) from September to January. Winter forage quality of bluebunch wheatgrass (*Pseudoroegneria spicata* [Pursh.] Á. Löve) was inadequate for maintaining nutrient needs of Rocky Mountain elk (*Cervus elaphus nelsoni* [Baily]).¹¹ An additional study in cattle reported weight loss in pregnant cows and calves grazing winter range^{12–14}; however, supplements have been used to offset that weight loss.^{15–17}

Anez-Osuna et al.¹⁸ observed decreases in values of CP from 10% to 7% and digestible energy from 12.5 to 10.8 Mcal/kg (5.7 to 4.9 Mcal/pound) DM in crested wheatgrass (*Agropyron cristatum* [L.] Gaertn.) from August to mid-October, resulting in the need to supplement backgrounded steers. Peng¹⁹ reported that meadow brome grass (*Bromus biebersteinii* [Roem. & Schult. {excluded}]) monocultures and mixtures with alfalfa (*Medicago sativa* L.), and cicer milkvetch

(*Astragalus cicer* L.) had the highest seasonal and total forage production in October. Baron²⁰ reported that stockpiled meadow brome grass provided high yield year to year with nutritive value suitable for cows through September and October and again in April.

The inclusion of a legume species in stockpiled pastures serves to increase the forage CP content and possibly reduce nitrogen inputs.²¹ However, due to the rapid decline in nutritive value after a heavy frost and yield loss during the winter, Baron²² concluded alfalfa was suitable for stockpiling only when grazed in September and October. Zobell²³ reported that on rangelands adding a semishrub such as forage kochia (*Bassia prostrata* [L.] A.J. Scott) to the seed mix increased forage biomass by 90% over the adjacent sagebrush range, and in addition CP ranged from 8% to 14% from August through March. Annual forages are also an option as a fall and winter forage providing high forage yield and nutritional quality; however, they require increased precipitation, land preparation, and seeding every year reducing economic sustainability.²⁴ The objectives of our study were to identify grasses and/or grass/legume mixtures on a landscape scale with increased dry-matter yield (DMY) and to determine the economic impact where the plots are grazed during the winter with no inputs other than raw manure from the cows and the use of a polyline electric fence.

Farm history and practice

Sieben Livestock Company Ranch is in west-central Montana in the Big Belt Mountains (47°05'06.81N, 111°31'19.81W; elevation 1,358 m [4,414 feet]). Nearly 20 years ago, the ranch made the switch to late spring calving to better match the cow's nutritional demand with the natural grass cycle. This management shift meant that cows during the winter were in the second trimester of pregnancy and could be maintained on stockpiled winter forage of lesser quality compared with cows in the final trimester or feeding a calf.⁸ This, resulted in reducing mechanically harvested hay from 1,361 kg/cow to 113 kg/cow (3,000 pounds/cow to 250 pounds/cow). With this change, the importance of winter forage, grazing efficiency, and forage production became more important.

Weight of the cattle in our study averaged about 544 kg (1,200 pounds) and was moderate-framed. Forage during the winter grazing period (December to April) is stockpiled during the growing year and grazed once during the winter for a 24-hour period of high-density, short duration grazing. Current grazing intensity includes about 1,000 cows with paddock densities ranging from 148 to 445 cows/ha (60 to 180 cows/acre), depending on forage production and grazing conditions. The ranch typically maintains 272 kg (600 pounds) of hay per cow in case of an emergency to help manage cow body condition.

Plant materials

Plant materials were selected based on their adaptation to annual precipitation and forage production. Legumes were included based on their adaptation and to determine their ability to potentially increase forage production and more importantly forage nutritional quality. Grass and grass-legume pastures of the following species: meadow brome grass (cv. Cache), intermediate wheatgrass (*Thinopyrum intermedium* [Host; Barkworth & D.R. Dewey; cv. Rush]), tall fescue (*Schedonorus arundinaceus* [Schreb.; Dumort.; cv. Fawn]), orchardgrass (*Dactylis glomerata* L. [cv. Paiute]), and crested wheatgrass (cv. Hycrest II) with the following legumes: alfalfa (cv. Ladak), sainfoin (*Onobrychis viciifolia* Scop. [cv. Eski]), and small burnet (*Sanguisorba minor* Scop. [cv. Delar]) were established for the purpose of evaluating their potential as a fall and winter forage along with their seeding rates (Table 1).

Site characteristics

The soil type for 10.6 ha (26 acres) of the 14.2 ha (35 acres) planted is a Monad Loam with 2% to 10% slope with the remaining 3.6 ha (9 acres) classified as Monad and Libeg soils, with a 25% to 45% slope.²⁵ The site was fallowed for 3 years before planting. In 2012, the site received one application of glyphosate (1.5 kg [1.4 pounds] of active ingredient/ha) before

Table 1. Grass and legume species used in our study along with the seeding rate (kg/ha), based on pure live seed.

Species/mixes	Seeding rate (kg/ha)
Meadow brome (MB) ^b	15
Intermediate wheatgrass (IWG)	15
Tall fescue (TF)	12
Crested wheatgrass (CWG)	12
Orchardgrass (OG)	12
MB + IWG + small burnet	7/7/2
MB + IWG + sainfoin	7/7/3
MB + IWG + alfalfa	7/7/2
MB + TF + alfalfa	7/5/2
MB + TF + sainfoin	7/5/3
OG + alfalfa	12/2
Control ^a	N/A

^a Located adjacent to the research plots.
^b Meadow brome (MB); intermediate WG (IWG); tall fescue (TF); orchardgrass (OG); and crested WG (CWG).



Figure 1. Seeding our study with a 1.5-m no-till drill on May 2013 at the Sieben Livestock Ranch near Cascade, Montana.

the May 13 planting and the remaining weeds were mowed. The seeding was done May 2013 with a six-row, 1.5-m (5 feet) Truax no-till drill (Fig. 1). Plot size was 24 m (80 feet) × 91 m (300 feet) arranged in a randomized complete block design with four replications. Throughout our study the plots received no irrigation or fertilizer, except for manure from grazing cows (~4,536 kg [10,000 pounds/year], based on 4.5 kg [10 pounds] per animal/day). Animals were turned in the day after the winter forage sampling, which was February 23

(1,009 cows), April 7 (1,117 cows), and March 8 (962 cows) for 2015, 2016, and 2017, respectively.

Site precipitation and temperature

Our study site, on the Sieben Livestock Ranch, is located outside of Cascade, Montana on Adel Road (47°05′06.81N, 111°31′19.81W) at 1,358 m (4,414 feet) above sea level. Long-term (1981–2010) average annual precipitation (AAP) was 37.8 cm (14.9 inches) with 65% coming during the growing season from April to September. As a percent of AAP, yearly site precipitation was 75%, 94%, 118%, 95%, 97%, and 107% in 2012 to 2017, respectively, during our study (Table 2). Average long-term snowfall is 112 cm (44 inches) throughout the winter. Long-term precipitation (1981–2010) distribution was 10%, 65%, and 25% of AAP occurring between January to April, May to September, and October to December, respectively.²⁶

Mean long-term minimum and maximum temperatures ranged from −9.6 °C (14.7°F; December) to 27.4 °C (81.3°F; July).²⁶ During 2012 to 2017, minimum temperatures ranged from −15.1 °C (4.8°F) to −8.2 °C (17.2°F) and maximum temperatures ranged from 26.2 °C (79.2°F) to 31.8 °C (89.2°F; Table 2). Four years during the 6-year study period were colder than the long-term minimum, and 4 years were hotter than 27.4 °C (81.3°F), the long-term maximum.

Stand establishment and persistence

Grass and legume seedling density was measured on September 10, 2013 in the grass monocultures and grass/legume mixtures. Grass and legume plant density was measured before each spring harvest date through 2017 in the grass monocultures and grass/legume mixtures. Both densities were taken using the grid system²⁷ in which the number of 12.7 cm² (5 inches²) squares in a 1 m² (1.1 yard²)

Table 2. Yearly (2012–2017) minimum and maximum temperatures and precipitation along with 30-year averages at the Sieben Livestock Ranch near Cascade, Montana.

Year	Minimum temperature °C	Maximum temperature °C	Precipitation (cm)
2012	−8.4	30.1	28.4
2013	−11.0	29.4	35.6
2014	−15.1	26.2	44.7
2015	−8.2	28.4	35.8
2016	−14.1	27.3	34.5
2017	−12.8	31.8	36.8
Long-term*	−9.6	27.4	37.8

* Long-term 30-year average for the site (1981–2010).



Figure 2. Example of taking plant frequency data on established pastures to estimate plant density. This is a 1 square meter frame containing 12.7 cm² squares at the Sieben Livestock Ranch near Cascade, Montana.

grid that contained one rooted plant of the target species were counted and converted to a density score of plants/m² (Fig. 2). We used generalized linear models (PROC GLM in SAS²⁸) with pasture species and mixtures and years as fixed effects and replication as a random effect to model grass and legume plant density and forage production. Unless otherwise noted, all mean separations were made based on Duncan's multiple range test at the 0.05 probability level as computed by PROC GLM.²⁸

During the establishment year (2013), all plots had combined grass and legume seedling densities above 132 plants/m² (12 plants/feet²; Table 3). Alfalfa established (2013) in grass mixtures (55–64 seedlings/m² [5–6 seedlings/feet²]) did significantly better ($P < 0.05$) than sainfoin (13–22 seedlings/m² [1.2–2.0 seedlings/feet²]) and small burnet at 10 seedlings/m² (1.0 seedlings/feet²; Table 3). Visual observations indicated that regardless of the grass species or mixtures, grass plant density remained high from 2014 to 2017. However, legume plant density declined after the initial establishment through 2016 in all treatments. Due to better initial establishment, alfalfa density declined more rapidly losing from 5 to 18 plants/m² (0.5–1.7 plants/feet²); whereas, small burnet lost 1 plant/m² (0.1 plant/feet²). Plant density of alfalfa, sainfoin, and small burnet increased by 30%, 18%, and 17%, respectively, in 2017 (Table 3).

October–November forage production

We estimated winter forage production using an October/November harvest. Within each plot, two 1.2 × 24.4 m (4 × 80 feet) subsamples were harvested to a 10 cm (4 inches) stubble height by a Swift Current sickle bar harvester (Swift Machining & Welding LTD, Swift Current, SK, Canada) on 11 October 2014 (Fig. 3). In 2015, 2016, and 2017, using the same protocol, sampling was done on October 27, October 25, and November 2, respectively. Subsamples (~250 g [~ 0.5 pound]) for DM weight calculations and forage nutritive value were taken from each plot and dried to a constant weight at 60 °C (140°F).

Fall forage production data by year and also a combined total across years showed that intermediate wheatgrass and intermediate wheatgrass/meadow brome mixtures consistently had the highest ($P < 0.05$) forage production regardless of yearly variation in temperature and precipitation between 2013 and 2017 when compared with tall fescue, orchardgrass, orchardgrass/alfalfa mixtures, and the native range (Table 4). Based on this data, the inclusion of either alfalfa, sainfoin, or small burnet in the different grass mixtures had little to no effect on fall forage production.

The initial forage production (i.e., DMY) in 2014 was higher in all species and mixtures, compared with 2015, 2016,

Table 3. Seedling and plant density (plants/m²) for grass and grass/legume mixtures from 2013–2017 at the Seiben Livestock Ranch near Cascade, Montana.

	Establishment 2013				2014		2015		2016		2017	
Species	Grass density		Legume density		Legume density		Legume density		Legume density		Legume density	
Meadow brome (MB)	142	cd*										
Intermediate WG (IWG)	151	abc										
Tall fescue (TF)	144	bcd										
Orchardgrass (OG)	160	a										
Crested WG (CWG)	132	d										
Grass/legume mixtures												
MB + IWG + alfalfa	145	bcd	55	a	38	a	33	a	9	a	26	ab
MB + IWG + sainfoin	144	bcd	22	b	5	b	0	b	0	b	7	b
MB + IWG + small burnet	144	bcd	10	b	12	b	7	b	1	b	6	b
MB + TF + alfalfa	144	bcd	64	a	48	a	45	a	5	b	43	ab
MB + TF + sainfoin	144	bcd	13	b	5	b	3	b	0	b	11	b
OG + alfalfa	156	ab	56	a	30	a	33	a	18	a	59	a

^a Density values with different letters in the same column are significantly different at $P < 0.05$.

^b Meadow brome (MB); wheatgrass (WG); intermediate WG (IWG); tall fescue (TF); orchardgrass (OG); and crested WG (CWG).



Figure 3. Example of mechanically harvesting forage in October to November to estimate available forage for winter grazing on our study plots at the Seiben Livestock Ranch near Cascade, Montana.

Table 4. Fall mean dry-matter yield (kg/ha) from Sieben Livestock Ranch near Cascade, Montana.

											Fall no. cows at 65% utilization (13 kg DM/cow day)
	Fall dry-matter yield (October-November)										
Monocultures	2014		2015		2016		2017		Comb.		
Meadow brome (MB)	4,598	abc*	2,407	cd	1,781	cd	1,399	cde	2,546	b	49
Intermediate WG (IWG)	4,923	ab	3,167	a	2,315	ab	2,833	ab	3,310	a	64
Tall fescue (TF)	3,156	d	1,636	d	1,105	ef	834	de	1,682	c	33
Orchardgrass (OG)	3,769	cd	1,165	e	929	f	572	e	1,609	c	31
Crested WG (CWG)	4,238	abc	2,810	ab	1,446	de	1494	cd	2,497	b	48
Grass/legume mixture											
MB + IWG + alfalfa	4,817	ab	2,891	a	2,518	a	3,039	a	3,317	a	64
MB + IWG + sainfoin	4,937	ab	2,923	a	2,215	abc	2,619	ab	3,174	a	61
MB + IWG + small burnet	5,129	a	3,205	a	2,352	ab	2,800	ab	3,372	a	65
MB + TF + alfalfa	4,239	abc	2,089	cd	2,190	abc	2,077	bc	2,649	b	51
MB + TF + sainfoin	4,255	abc	1,795	d	1,972	bc	2,047	bc	2,517	b	49
OG + alfalfa	4,030	bcd	1,167	e	1,046	ef	1,003	de	1,811	c	35
Native range	.		951	e	710	f	1,608	cd	1,090	d	21

^a Dry-matter yield values with different letters in the same column are significantly different at $P < 0.05$.

^b Meadow brome (MB); wheatgrass (WG); intermediate WG (IWG); tall fescue (TF); orchardgrass (OG); and crested WG (CWG).

Table 5. Mean winter forage crude protein, acid detergent fiber, digestible neutral detergent fiber, water-soluble carbohydrates, total digestible nutrients, and relative feed quality from the Sieben Livestock ranch near Cascade, Montana during 2015–2016.

Species [^]	Crude protein [†]			Acid detergent fiber [‡]			Digestible neutral detergent fiber [§]			Water-soluble carbohydrates			Total digestible nutrients [¶]		Relative forage quality [#]	
	%		kg/ha	%		kg/ha	%		kg/ha	%		kg/ha	%		%	
Meadow brome (MB)	6.1	bc*	155	57.6	e	1467	31.6	de	805	0.0	d	0	36.9	e	31.1	e
Intermediate WG (IWG)	7.0	abc	232	49.6	e	1642	35.1	bcde	1162	1.8	bc	60	46.0	b	53.1	abc
Tall fescue (TF)	8.3	a	140	50.9	cde	856	37.7	abc	634	3.7	a	62	44.7	bc	53.9	abc
Orchardgrass (OG)	7.0	abc	113	50.2	de	808	39.2	ab	631	1.8	bc	29	45.2	bc	61.2	a
Crested WG (CWG)	6.0	bc	150	51.5	bcde	1286	30.6	e	764	1.5	bcd	37	44.1	bc	44.0	bcde
MB + IWG + alfalfa	7.5	ab	249	52.8	bcde	1158	34.9	bcde	1158	1.3	bcd	43	42.4	cd	47.8	abcd
MB + IWG + sainfoin	7.1	abc	225	53.5	cd	1698	32.0	de	1016	0.7	cd	22	42.0	cd	43.7	bcde
MB + IWG + small burnet	5.6	bc	189	53.9	b	1818	36.0	bcd	1214	0.8	cd	27	40.9	d	38.3	cde
MB + TF + alfalfa	7.0	abc	185	57.0	a	1510	33.6	cde	890	0.8	cd	21	37.4	e	34.0	de
MB + TF + sainfoin	7.1	abc	179	57.1	a	1438	31.9	de	803	0.5	cd	13	37.5	e	36.1	de
OG + alfalfa	6.9	abc	125	50.7	cde	918	42.1	a	763	2.6	ab	47	44.7	bc	57.7	ab
Native range	5.3	c	58	46.8	f	510	33.0	cde	360	1.5	bcd	16	49.0	a	47.5	abcd

* Different letters in the same column are significantly different at $P < 0.05$.[†] Crude protein (CP) is estimated by multiplying the nitrogen concentration in the forage (N%) by 6.25.[‡] Acid detergent fiber is the residue remaining after boiling a forage sample in an acid detergent solution. It represents the cellulose, lignin, and silica values in the forage.[§] Digestible neutral detergent fiber (dNDF) is the portion of neutral detergent fiber digested by animals.^{||} Water-soluble carbohydrates are a measure of water-soluble sugars, including simple sugars and fructans.[¶] Total digestible nutrients are the sum of CP, fat (multiplied by 2.25), nonstructural carbohydrates, and dNDF.[#] Relative forage quality is an estimate of how much available energy a nonlactating animal will obtain daily from a forage if that is all it is fed.[^] Meadow brome (MB); wheatgrass (WG); intermediate WG (IWG); tall fescue (TF); orchardgrass (OG); and crested WG (CWG).



Figure 4. Cows (1,117) grazing 7 April 2016 for 24 hours on our study site at the Sieben Livestock Ranch, near Cascade, Montana.

and 2017 (Table 4), which likely resulted from stockpiled soil nutrients since our study site was fallowed for 3 years before planting. After the initial flush of available nutrients in 2014, fall forage production in the following 3 years remained relatively constant (Table 4).

Forage nutritive value

Dried forage samples used for DMY determination were ground to pass through a 1-mm (0.04 inches) screen using a Wiley mill grinder before analysis. Ground plant samples were scanned with a near infrared reflectance spectroscopy (NIRS) instrument (Model 6500; Pacific Scientific Instruments, Silver Spring, MD).²⁹ Wavelength scans from the NIRS were then used to predict CP, acid detergent fiber (ADF), digestible neutral detergent fiber (dNDF48), water-soluble carbohydrates (WSC), total digestible nutrients, and relative feed quality using the NIRS consortium equation.²⁹

Winter stockpiled forage CP concentrations for the grass monocultures and grass/legume mixtures showed that plant materials with CP concentrations $\geq 7\%$, recommended as a minimum for maintaining nonlactating cows in mid-to-late gestation,⁹ were intermediate wheatgrass, tall fescue, orchardgrass, meadow brome/intermediate wheatgrass mixtures with alfalfa and sainfoin, and meadow brome-tall fescue mixtures with sainfoin and alfalfa (Table 5). However, due to reduced forage production in tall fescue, orchardgrass, orchardgrass/alfalfa mixture, and native range, total CP (kg/ha) produced was less. Total CP ranged from 58 to 249 kg/ha (51.6–222.1 pounds/ac) for native range and meadow brome-intermediate wheatgrass-alfalfa mix, respectively (Table 5). Grasses and mixtures with the highest CP values also exhibited the lowest ADF values with the exception of the native range, which had the lowest CP and ADF values (Table 5). Orchardgrass and

orchardgrass-alfalfa had the highest percentage of dNDF followed by tall fescue, meadow brome-intermediate wheatgrass-small burnet, and intermediate wheatgrass (Table 5). In general, WSC values were low in stockpiled forage with tall fescue and orchardgrass-alfalfa having WSC values greater than the rest. Total digestible nutrient values were greater ($P < 0.05$) in forage collected from the native range than all other grasses and grass-legume mixtures. Relative feed quality percentages ranged from 31.1% (meadow brome) to 61.2% (orchardgrass) with orchardgrass-alfalfa, tall fescue, intermediate wheatgrass, and meadow brome-intermediate wheatgrass-alfalfa having similar values (Table 5).

Potential animal carrying capacity

The potential animal carrying capacity, which ranged from 21 cows/day in the native range to > 60 cows/day in the intermediate wheatgrass and intermediate wheatgrass/meadow brome mixtures with small burnet, alfalfa, and sainfoin, was based on a 4-year average forage production and assumptions of 65% forage utilization by the animal, and 13.6 kg (30 pounds) of DM intake/day/cow (Table 4, Fig. 4).

Economic impact and implication of improved winter pastures

As producers consider implementing the use of improved dryland winter pastures as a main source of forage, they may question if they can expect a positive net financial impact by improving their forage base. To answer this question, we used a partial budget analysis that considered both increased and decreased revenue as well as increased and decreased expenses. Assumptions used in the partial budget analysis included

Table 6. Annual net financial impact, total establishment costs, payback period, and the internal rate of return (IRR) of improved forage varieties for our Seiben Livestock Ranch study near Cascade, Montana.

Forage [^]	Annual net financial impact [*]	Total establishment costs [†]	Total establishment costs per ha [‡]	Payback period (y) [§]	Annual net financial impact per ha	IRR [¶]
Intermediate (IWG)	\$147,505	\$114,474	\$181.14	0.78	\$233.44	129%
MB + TF + alfalfa	\$139,735	\$110,297	\$139.74	0.79	\$177.04	127%
MB + IWG + alfalfa	\$147,580	\$119,308	\$189.17	0.81	\$234.01	124%
MB + IWG + sainfoin	\$146,170	\$126,898	\$192.51	0.87	\$221.75	115%
MB + TF + sainfoin	\$137,695	\$118,858	\$143.08	0.86	\$165.77	116%
TF	\$117,295	\$101,225	\$81.41	0.86	\$94.25	116%
MB + IWG + small burnet	\$121,140	\$114,912	\$185.22	0.95	\$195.25	105%
CWG	\$118,100	\$137,720	\$165.45	1.17	\$141.03	86%
MB	\$120,835	\$148,756	\$181.14	1.23	\$147.16	81%
OG + Alfalfa	\$119,735	\$204,487	\$177.06	1.71	\$103.68	58%
OG	\$114,520	\$206,012	\$158.53	1.80	\$88.14	56%

^{*} Annual net financial impact is the sum of the positive and negative changes within the partial budget of each forage variety and is calculated for 1,000 cows grazed for 100 days.

[†] The total establishment costs for each forage variety are considered the initial cost of investment and are equal to the establishment costs (\$/ha) multiplied by the total amount of hectares needed to produce the required forage.

[‡] Total establishment costs per hectare includes fixed costs and variable seed costs. Fixed costs are \$23.58/ha based on a 150 hp tractor using a 4.5 m foot drill. Variable seed costs ranged from \$0.89 kg⁻¹ to \$2.16 kg⁻¹ and were seeded according to the rates described in Table 1.

[§] The payback period is calculated as the forage total establishment cost divided by the annual net financial impact.

^{||} Annual net financial impact per hectare is equal to the annual net financial impact divided by the total hectares required for each forage variety to produce the amount of forage necessary to feed 1,000 cows over 100 days.

[¶] IRR is the internal rate of return for the investment into each forage variety and is equal to the discount rate that yields a net present value of \$0. The IRR is calculated over a period of 15 years after which time it was assumed that each forage would require re-establishment.

[^] Meadow brome (MB); wheatgrass (WG); intermediate WG (IWG); tall fescue (TF); orchardgrass (OG); and crested WG (CWG).

1,000 cows fed over winter, a 100-day winter grazing period, with an average cow intake of 13.6 kg (30 pounds) of DM/day for a total forage requirement of 1,360 Mg (1,500 tons). We assumed that alfalfa cost \$143.30/Mg (\$130/ton). With a hay savings of 1,247 kg (2,750 pounds)/cow/winter (1,361 to 113 kg [3,000 to 250 pounds]). This resulted in an average cost savings of \$178,750 compared with fed hay.

To estimate the increased expenses in the partial budget, total forage value was determined by using the following equation:

$$FV_i = \frac{GR}{(DMY_i * HE_i)} (2000) \quad (1)$$

Where FV_i is the calculated forage value for the i th variety (\$/Mg), GR is the assumed grazing rate (\$/ha), DMY_i is the dry-matter yield (kg/ha) for the i th variety, and HE_i is the

assumed harvest efficiency percentage for the i th variety. We assumed a grazing rate of \$49.40/ha (\$20/ac) and a harvest efficiency (forage utilization) of 65%. Assuming 7% CP as the minimum requirement for nonlactating cows in mid-to-late gestation,⁹ we know that of the 1,360 Mg (1,500 tons) of forage provided, 95 Mg (105 tons) of protein should be made available. We estimated the needed protein supplement to bring CP in the diet above 7% CP by taking the difference of the required animal intake (DM) and that provided by the stockpiled forage crop divided by 0.3 (assuming 30% protein block). We used a value of \$424.40/ha (\$385/ac) for the protein blocks. We determined the net annual financial impact of each forage variety that was calculated as the sum of the feed savings and total increased expenses (Table 6).

Frequently, producers want to know how long it will take to pay back their initial investment. The payback period was calculated as the initial cost of the investment divided by the

annual expected increase to cash flow. The establishment costs can be thought of as the initial cost of investment, which includes seed costs, and expenses associated with a 150 hp tractor and a 15-foot no-till drill (Table 6). Once this initial cost has been covered (pay back), the net annual financial impact is the benefit producers receive annually over the traditional feeding of alfalfa during the fall and winter (Table 6).

Conclusions

With ever-increasing operational costs of cow/calf production, profit margins continue to decrease, resulting in producers exploring alternative management practices to reduce costs. One such option, is to replace mechanically harvested forage with stockpiled forage to be grazed during the fall and winter. This may often require changes in management to include calving and breed back dates, along with intensive management practices. Based on the above assumptions and the Sieben Land and Livestock management program, our data support the renovation of pastures to improved grass, grass mixtures for stockpiled winter forage production, but also suggests that the economic return more than offsets the initial cost of renovation. Furthermore, the only fertilizer added to the plots from 2014 to 2017 was supplied by the grazing cows, suggesting even more cost savings to the producer. It is important to note that we have assumed no acreage restriction on the calculations (Table 6). This means we calculated the needed acreage based on the DMY of the chosen forage variety and assume the acreage is available through ownership or renting of additional land. If grazing land is unavailable or infeasible to rent, then the analysis must be constrained to the owned acreage and the alfalfa savings may decrease if the chosen forage variety requires more acres to be planted than are available. Based on payback periods, intermediate wheat grass (var. Rush) would be the optimal choice (Table 6). If the varieties were ranked according to net financial impact, meadow brome (var. Cache)-intermediate wheatgrass (var. Rush)-alfalfa (var. Ladak) would be the better choice. Discrepancies in ranking order by payback period and net financial impact result from the differences in seed costs. Intermediate wheatgrass, as well as many of the meadow brome mixtures with intermediate wheatgrass and tall fescue (var. Fawn), all have similarly low payback periods (Table 6). Thus, in making the decision of what variety to seed, producers should consider carefully seed availability and cost as well as the individual needs of their operations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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